

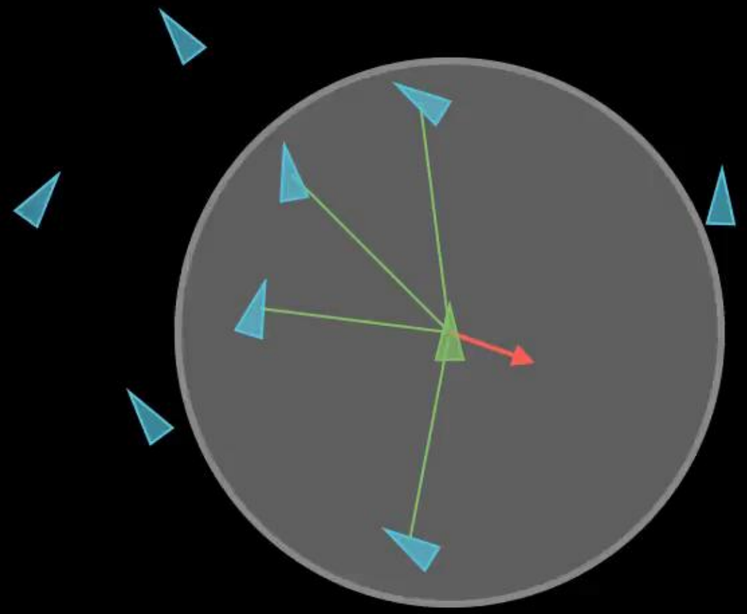




Swarm Intelligence

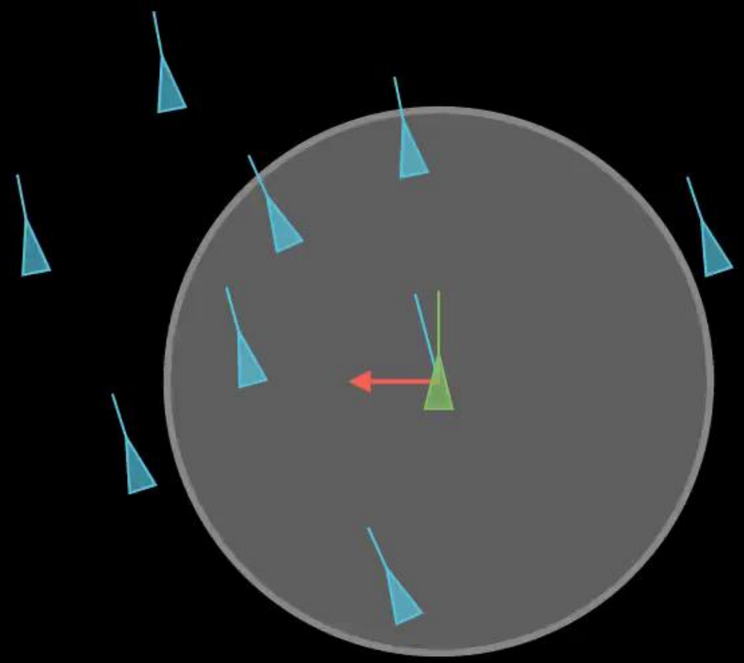
or

HOW TO GET A COOL TATOO IN YOUR SHELL IF YOU ARE A SEA SNAIL



Rules

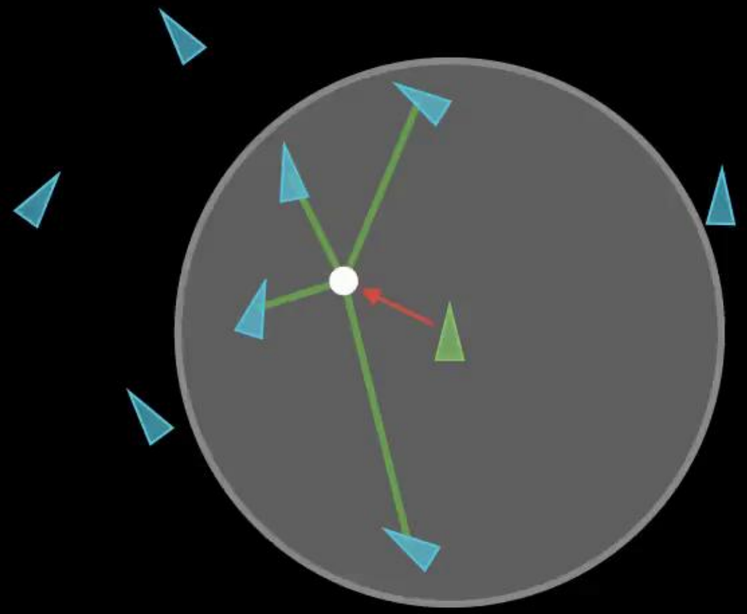
Separation



Rules

Separation

Alignment



Rules

- Separation
- Alignment
- Cohesion

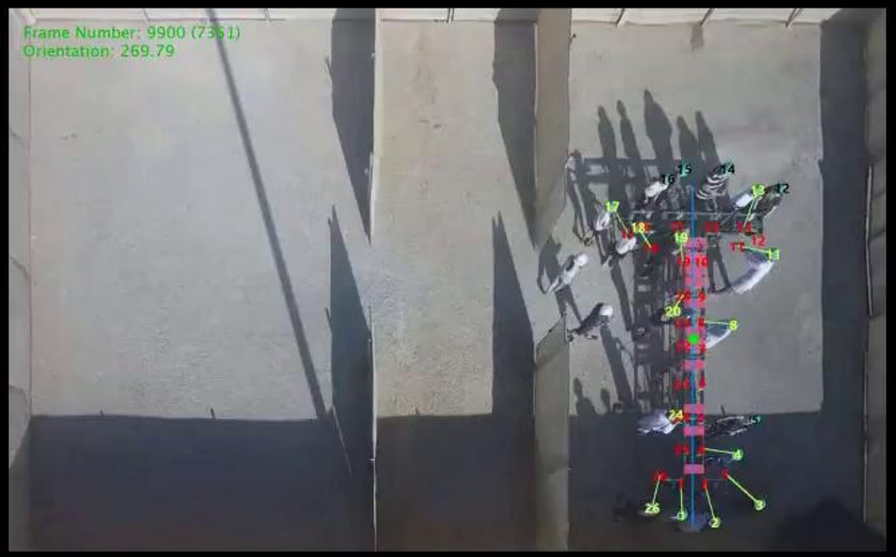
Florida Museum photo by Zachary Randall

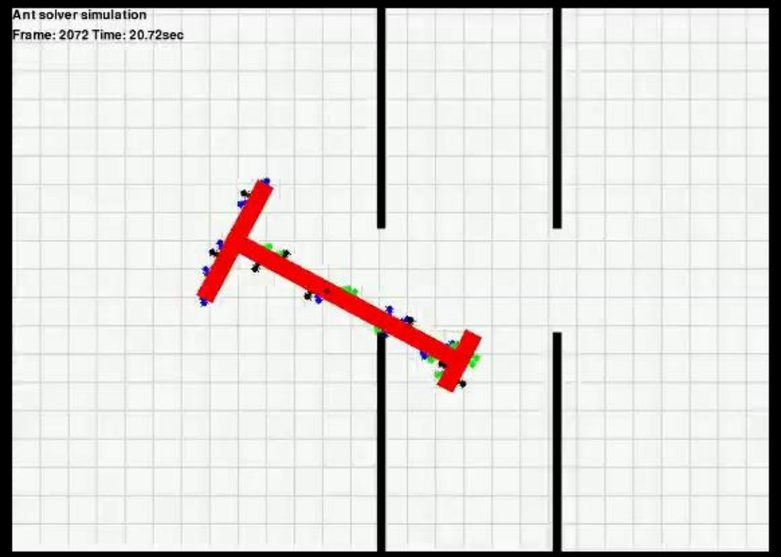


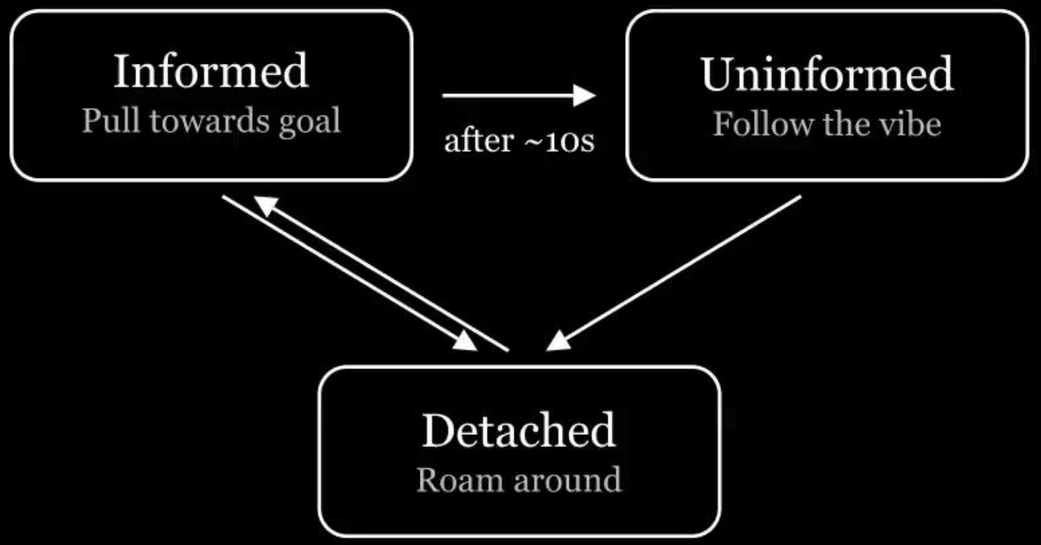


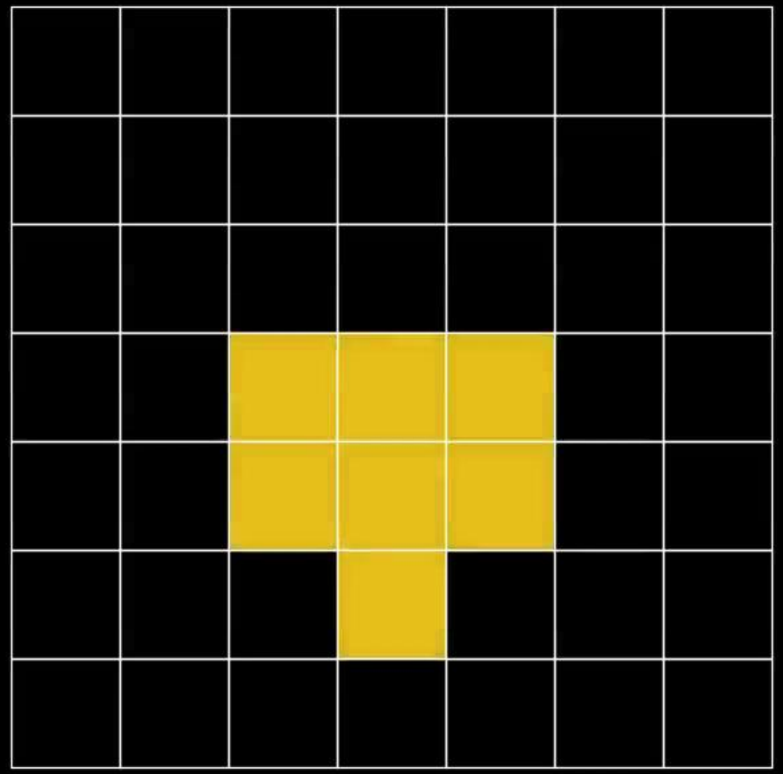

Frame Number: 4051 (3482)
Orientation: 88.40
Orientation (fixed): 88.40

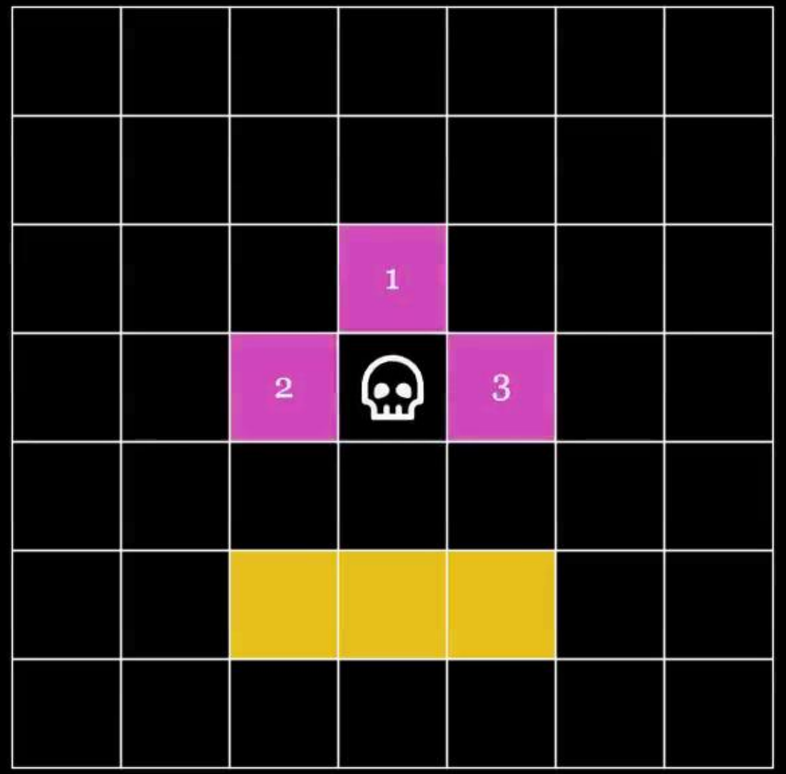



Frame Number: 9900 (7351)
Orientation: 269.79


Ant solver simulation
Frame: 2072 Time: 20.72sec

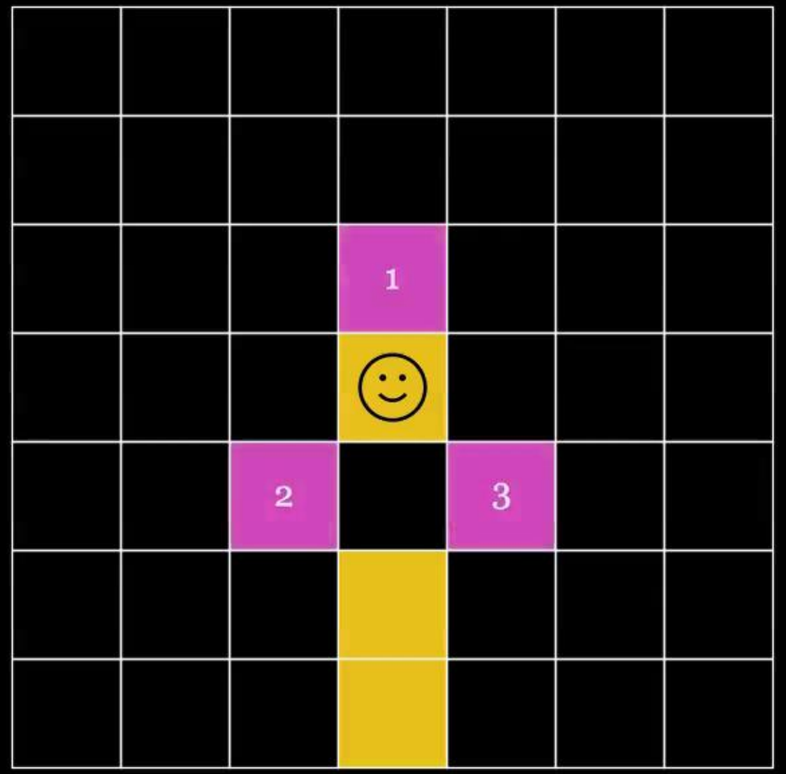

Informed
Pull towards goal
after ~10s
Uninformed
Follow the vibe
Detached
Roam around





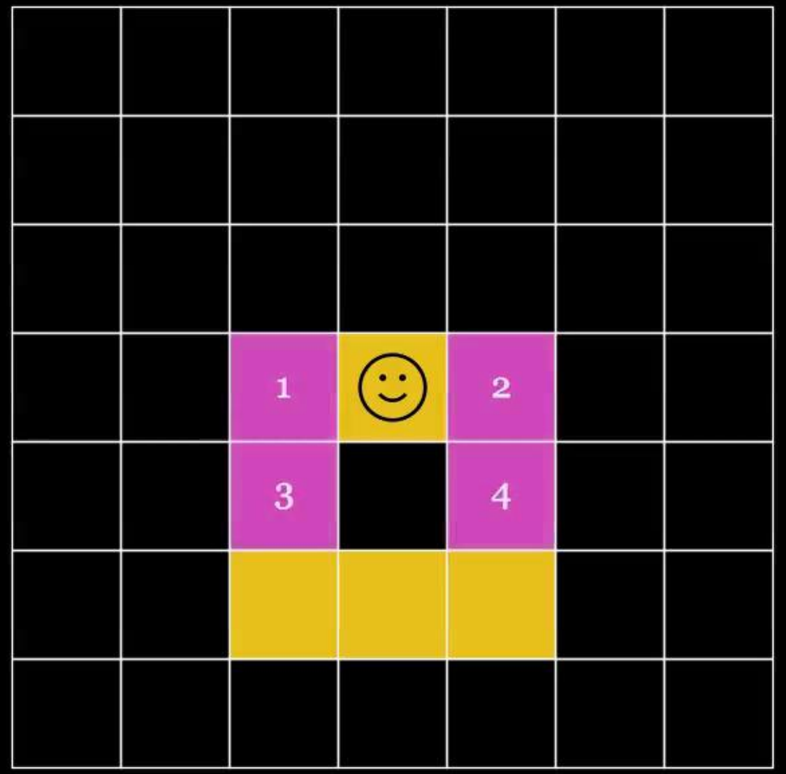
Rules

1. Any dead cell with three alive neighbors becomes alive



Rules

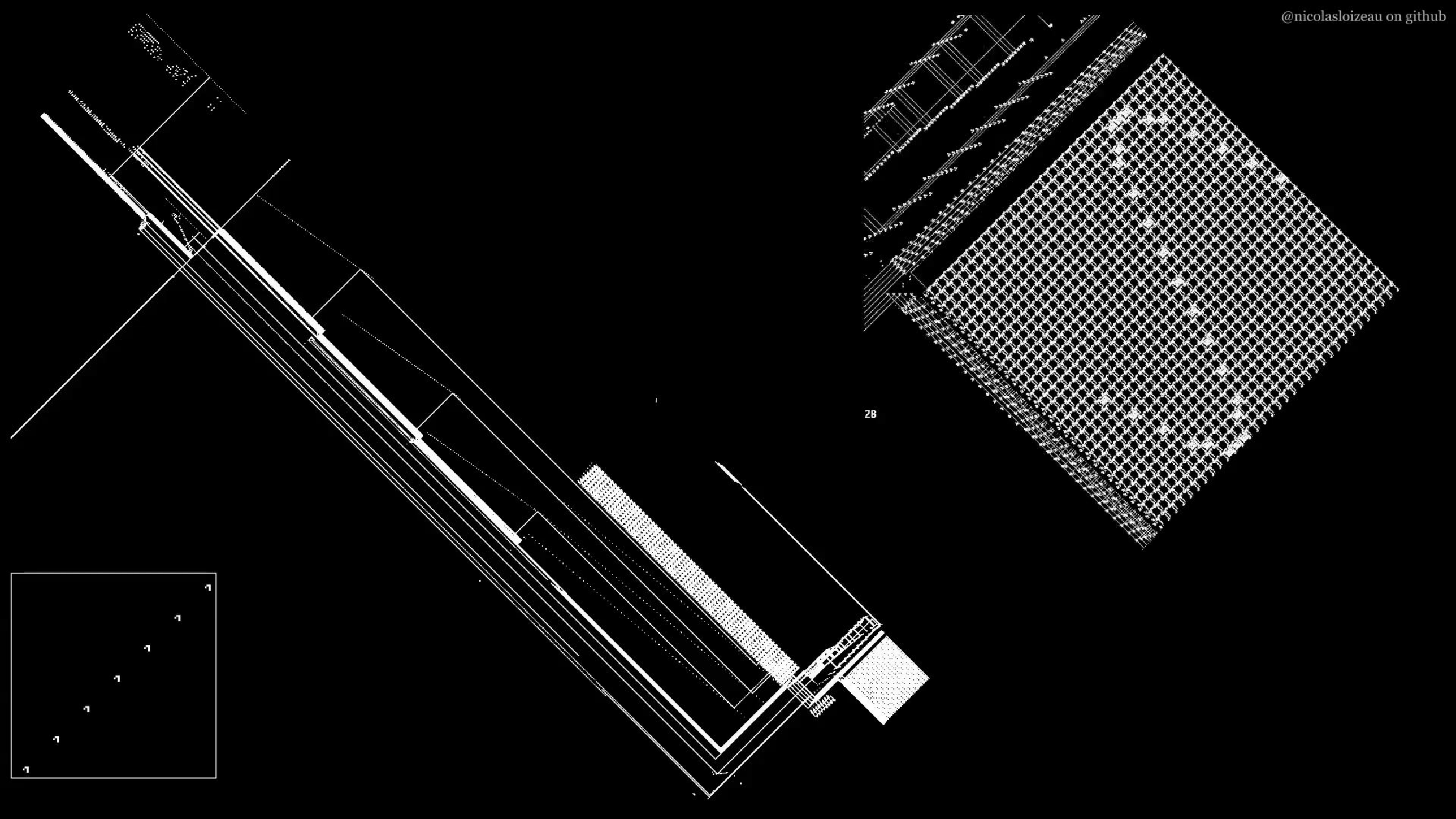
1. Any dead cell with three alive neighbors becomes alive
2. Any alive cell with two or three neighbors survives



Rules

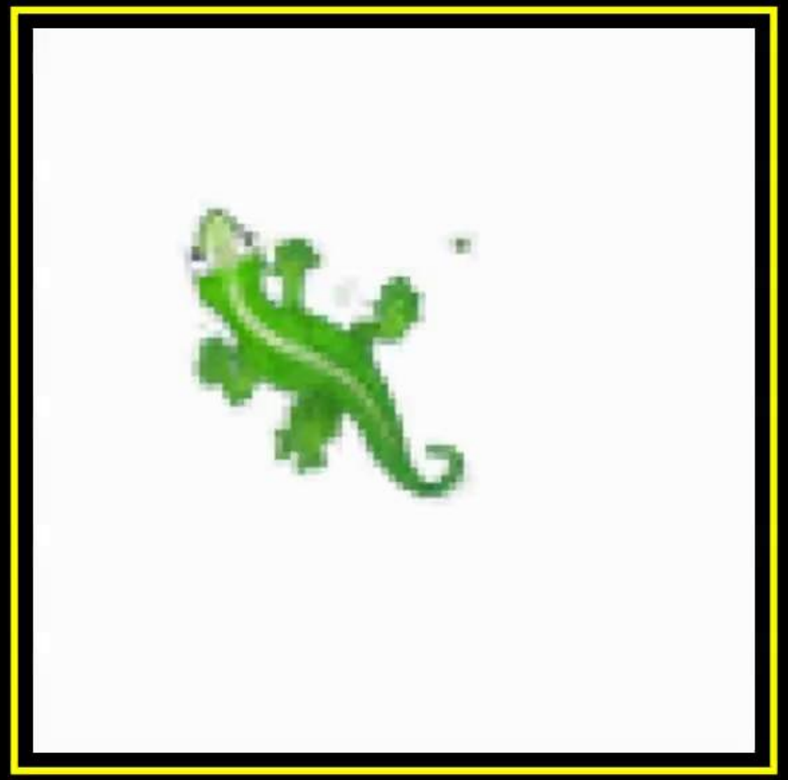
1. Any dead cell with three alive neighbors becomes alive
2. Any alive cell with two or three neighbors survives

@nicolasloizeau on github



$$(A, \mathbb{Z}^d, X, \tau)$$

Growing NCA (Mordvintsev et al., 2020, Distill)



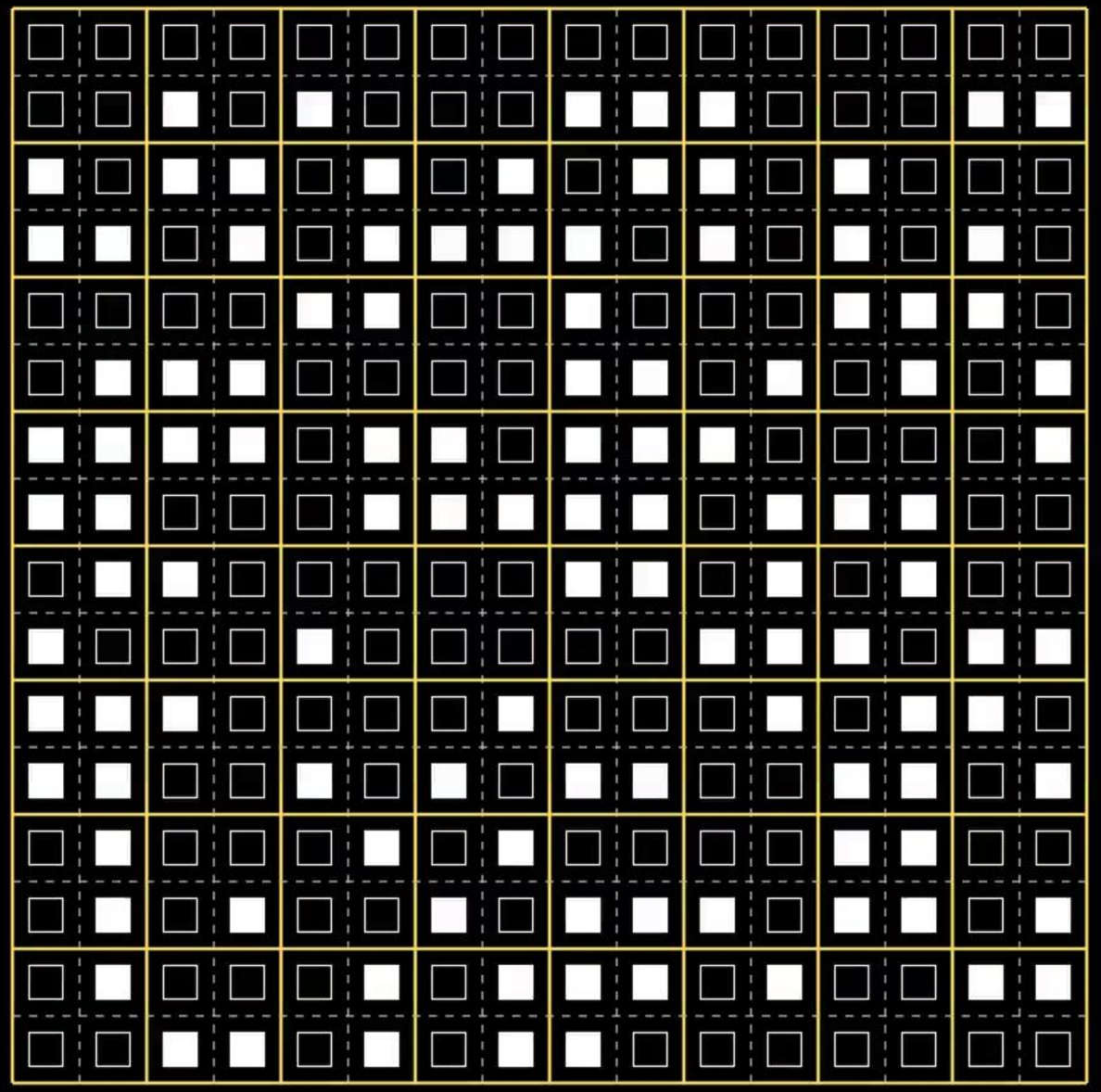
$$(A, \mathbb{Z}^d, X, \tau)$$



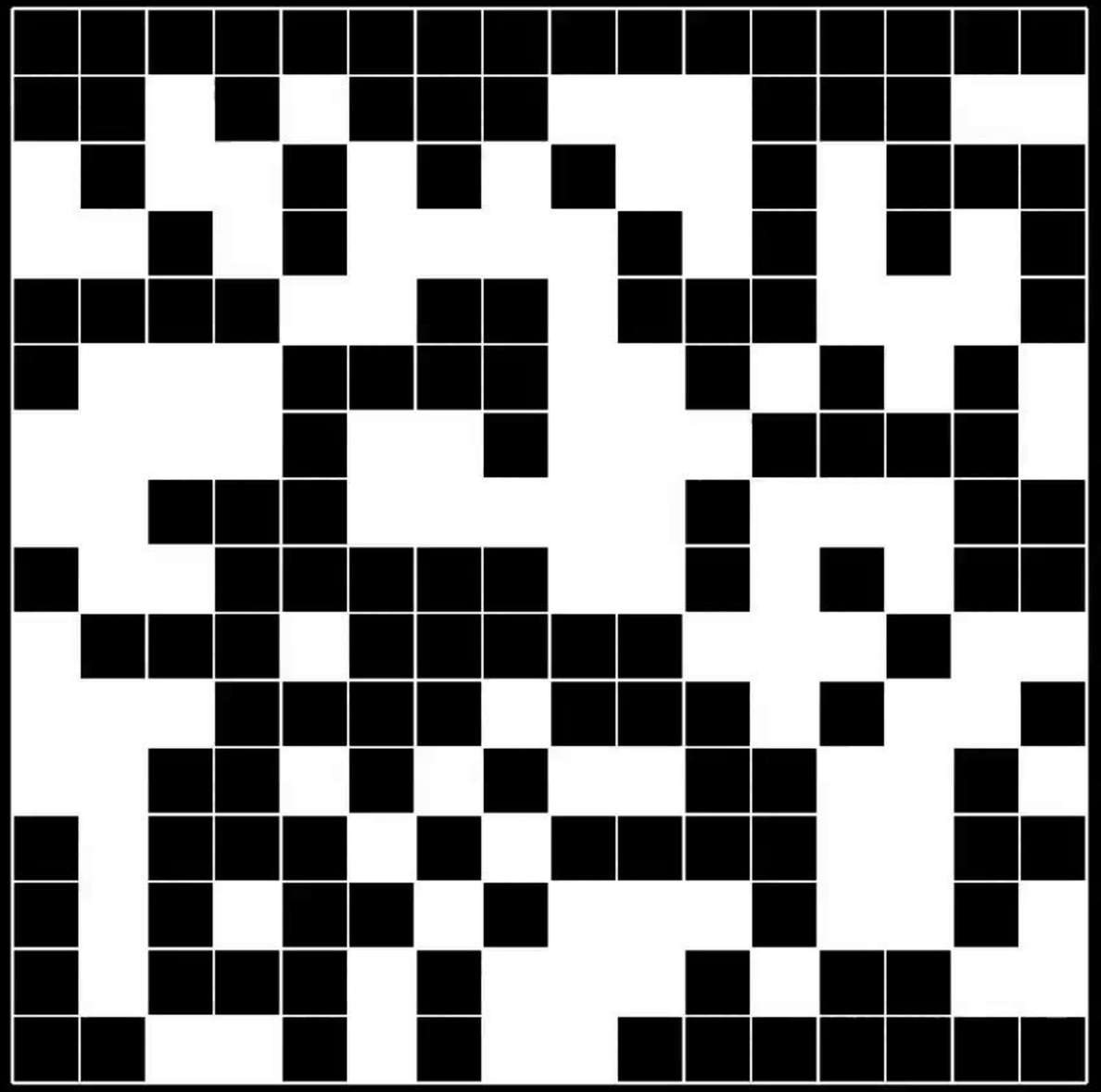
$$A = \mathbb{R}^{16}$$

$$\begin{bmatrix} r \\ g \\ b \\ a \\ h_0 \\ h_1 \\ h_2 \\ h_3 \\ h_4 \\ h_5 \\ h_6 \\ h_7 \\ h_8 \\ h_9 \\ h_{10} \\ h_{11} \end{bmatrix}$$

$$(A, \mathbb{Z}^d, X, \tau)$$



$(A, \mathbb{Z}^d, X, \tau)$



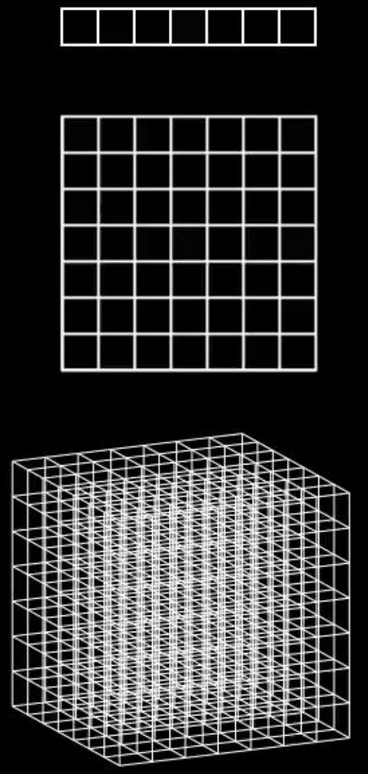
$(\mathbb{B}, -, -, -)$

$(A, \mathbb{Z}^d, X, \tau)$

$\mathbb{Z}^1$

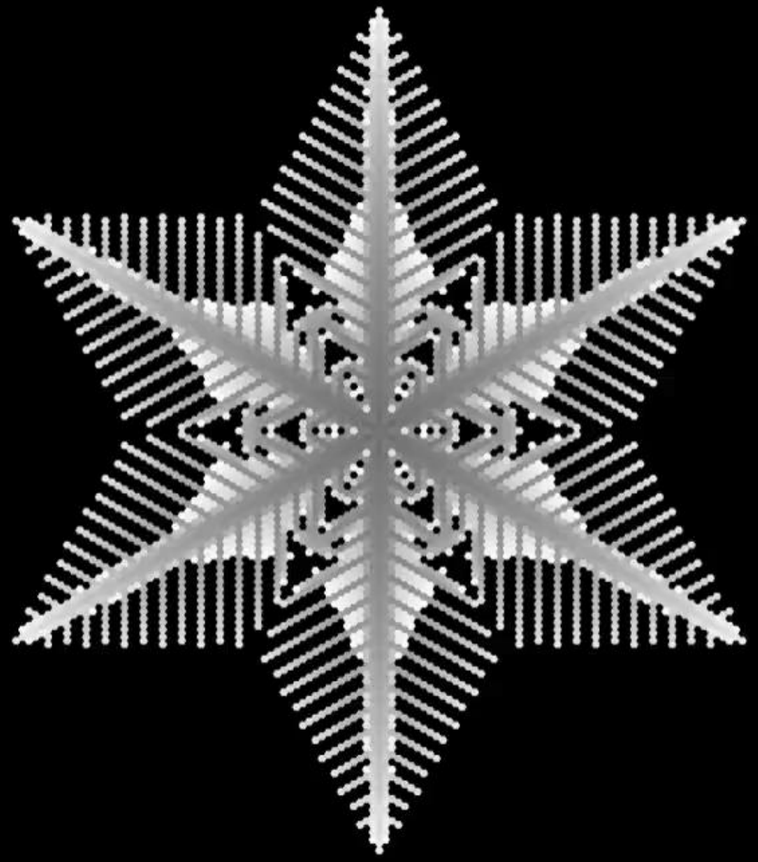
$\mathbb{Z}^2$

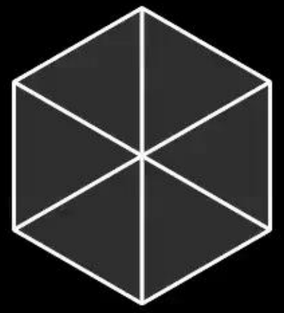
$\mathbb{Z}^3$



$(\mathbb{B}, \mathbb{Z}^1, _, _)$

$(A, \mathbb{Z}^d, X, \tau)$

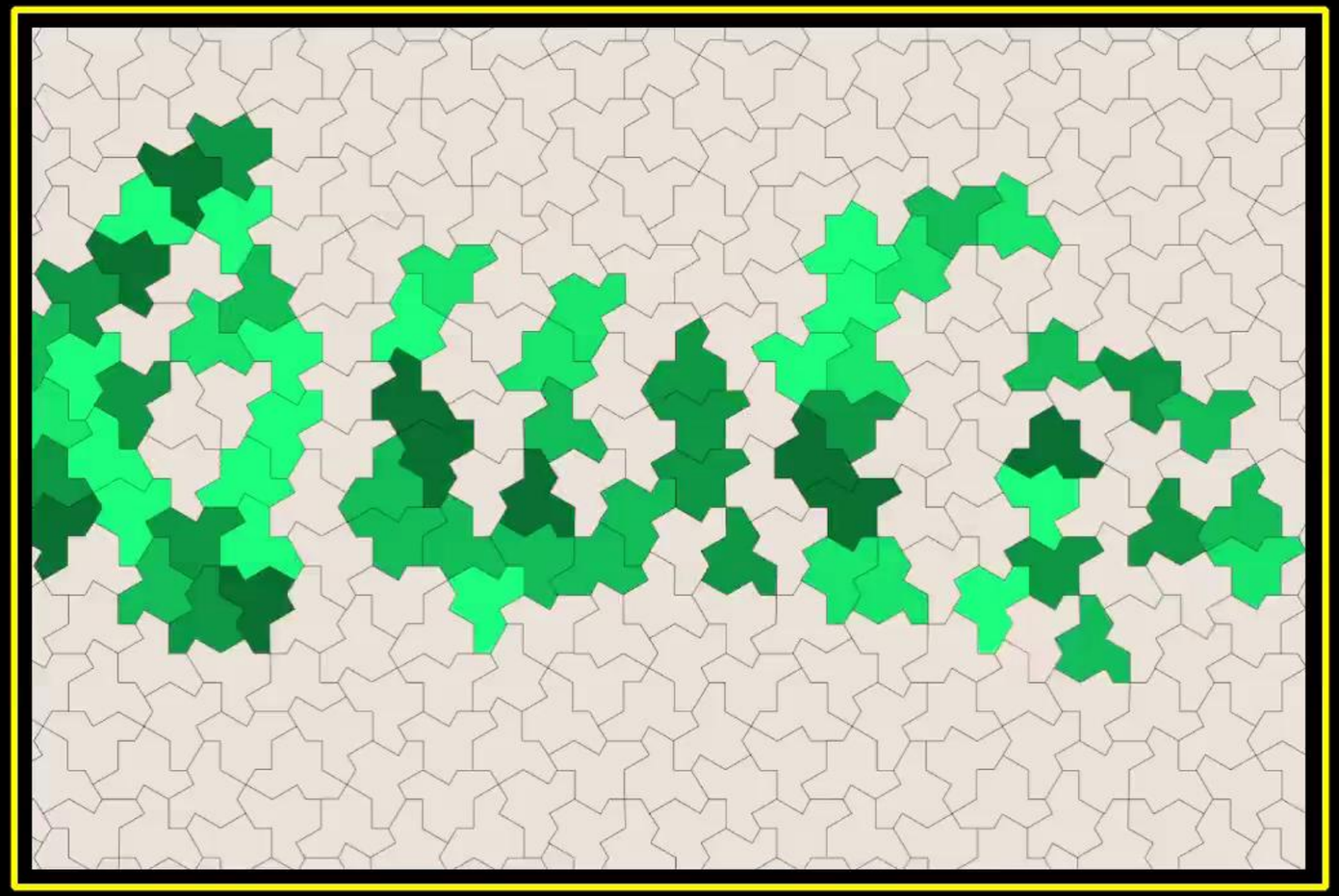




$(\mathbb{B}, \mathbb{Z}^1, _, _)$

$(A, \mathbb{Z}^d, X, \tau)$

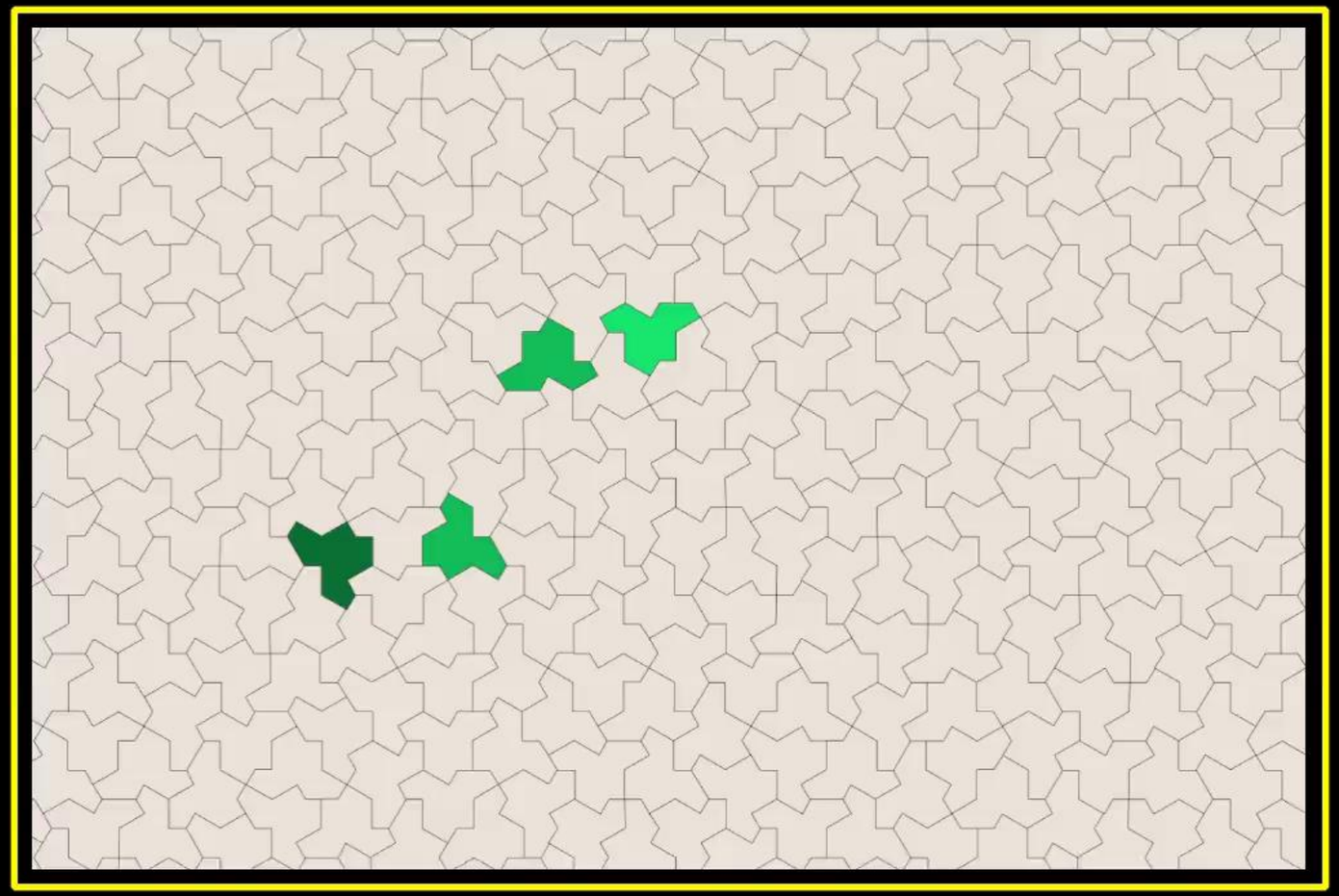
@henrypdonahue on Github



$(\mathbb{B}, \mathbb{Z}^1, _, _)$

$(A, \mathbb{Z}^d, X, \tau)$

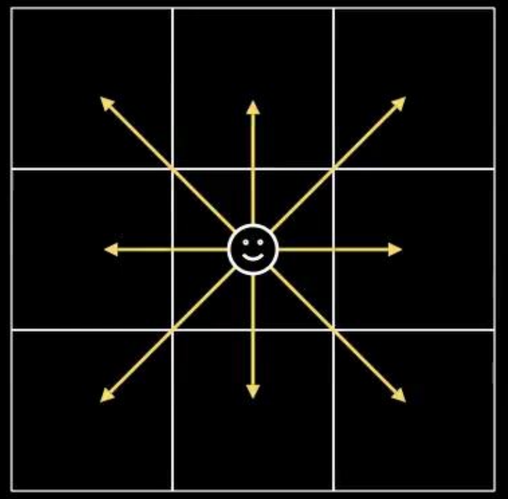
@henrypdonahue on Github



$(\mathbb{B}, \mathbb{Z}^1, _, _)$

$(A, \mathbb{Z}^d, X, \tau)$

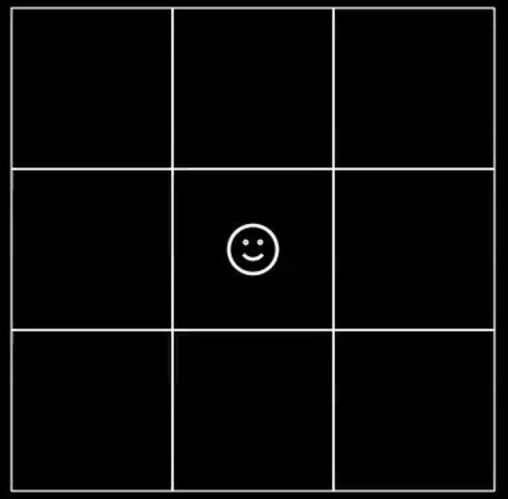
$$X =$$



$(\mathbb{B}, \mathbb{Z}^1, _, _)$

$(A, \mathbb{Z}^d, X, \tau)$

$$X = (\nwarrow, \uparrow, \nearrow, \leftarrow, \rightarrow, \swarrow, \downarrow, \searrow, \odot)$$



$(\mathbb{B}, \mathbb{Z}^1, _, _)$

$(A, \mathbb{Z}^d, X, \tau)$

Scope 1



Scope 2



Scope 3



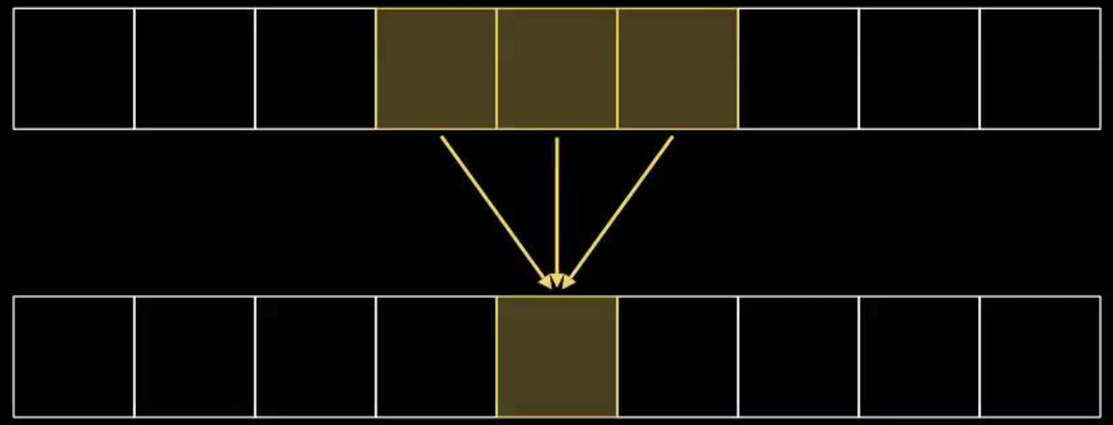
Scope 4



$(\mathbb{B}, \mathbb{Z}^1, _, _)$

$(A, \mathbb{Z}^d, X, \tau)$

$$\sigma : A^{|X|} \to A$$



$(\mathbb{B}, \mathbb{Z}^1, Scope3, _)$

$(A, \mathbb{Z}^d, X, \tau)$

$$\sigma : A^{|X|} \to A$$

$$\tau : C \to C$$



$(\mathbb{B}, \mathbb{Z}^1, Scope3, _)$

$(A, \mathbb{Z}^d, X, \tau)$

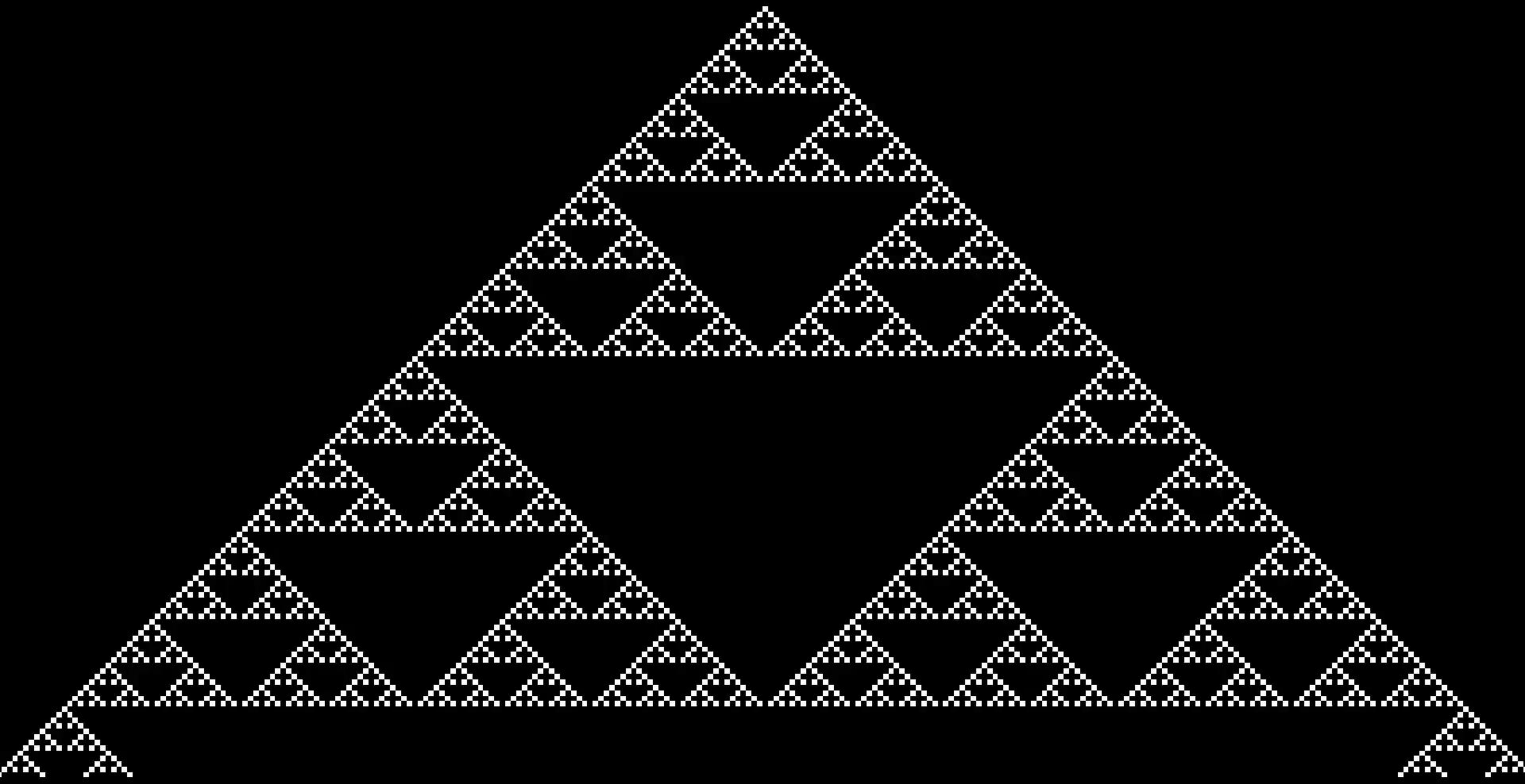
N	σ	σ_{90}	σ_{110}
000	a	0	0
001	b	1	1
010	c	0	1
011	d	1	1
100	e	1	0
101	f	0	1
110	g	1	1
111	h	0	0

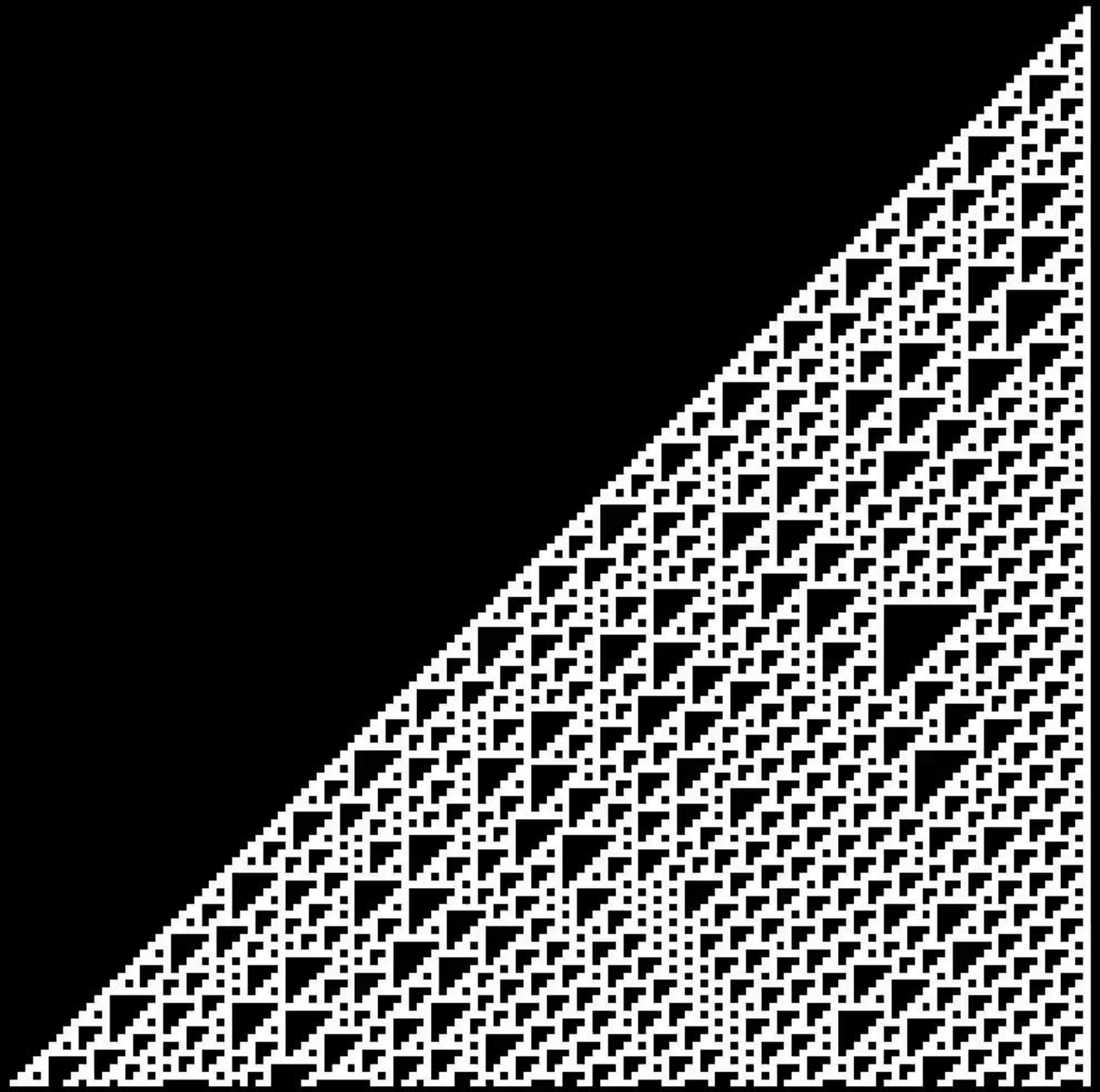
$hgfedcba \rightarrow$ Wolfram Scheme

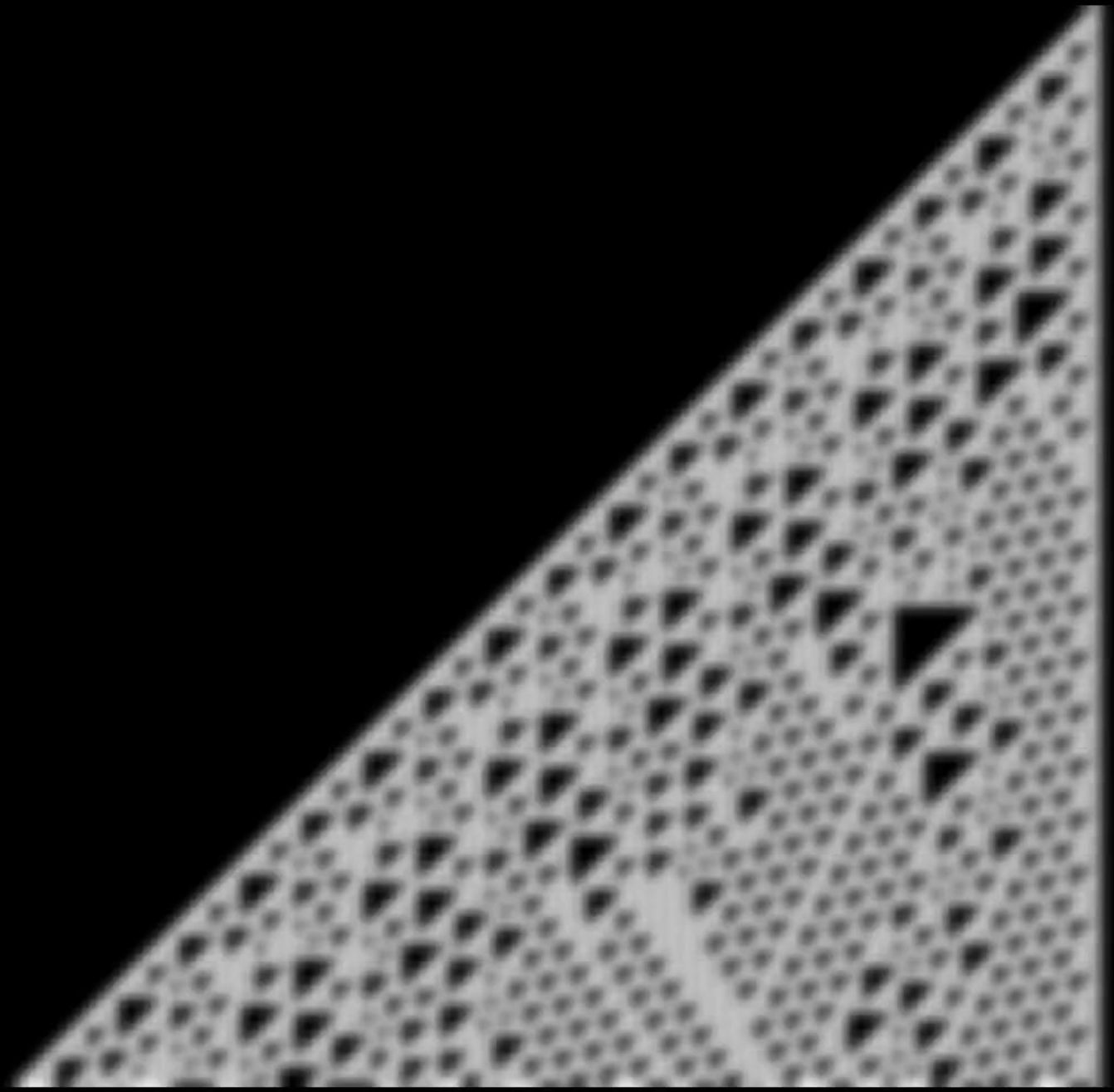
$01011010 \rightarrow 90$

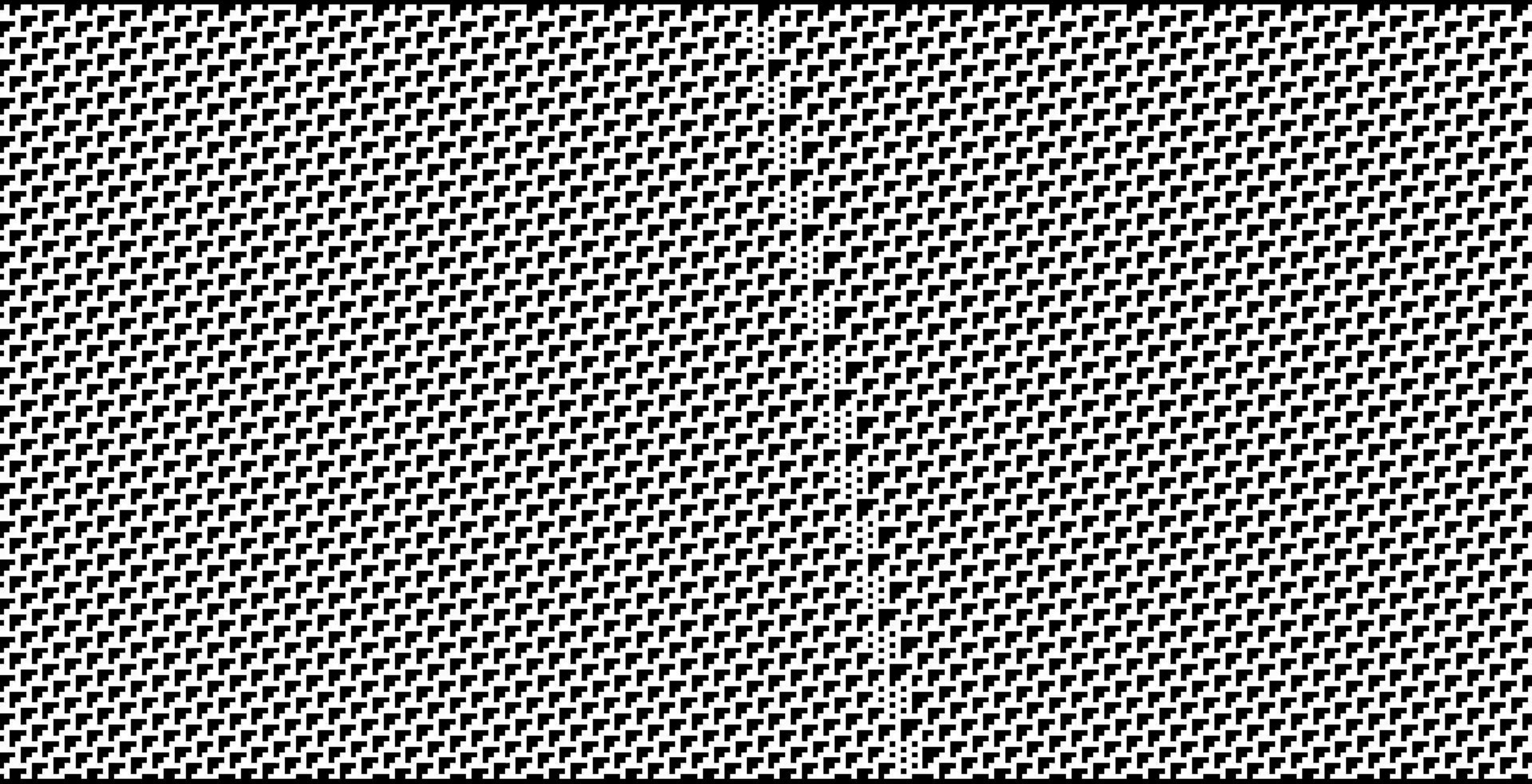
$01101110 \rightarrow 110$

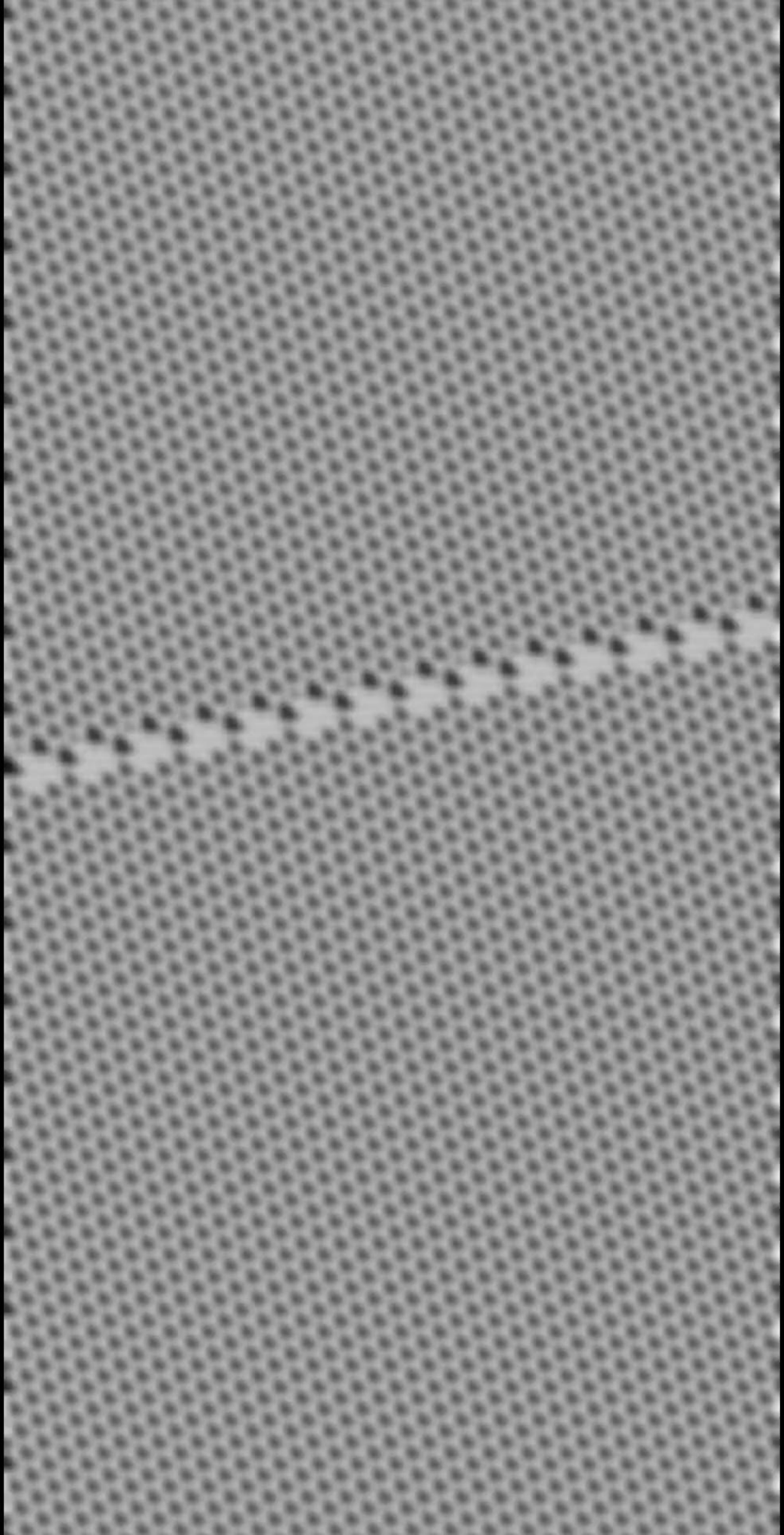
$(\mathbb{B}, \mathbb{Z}^1, Scope3, _)$

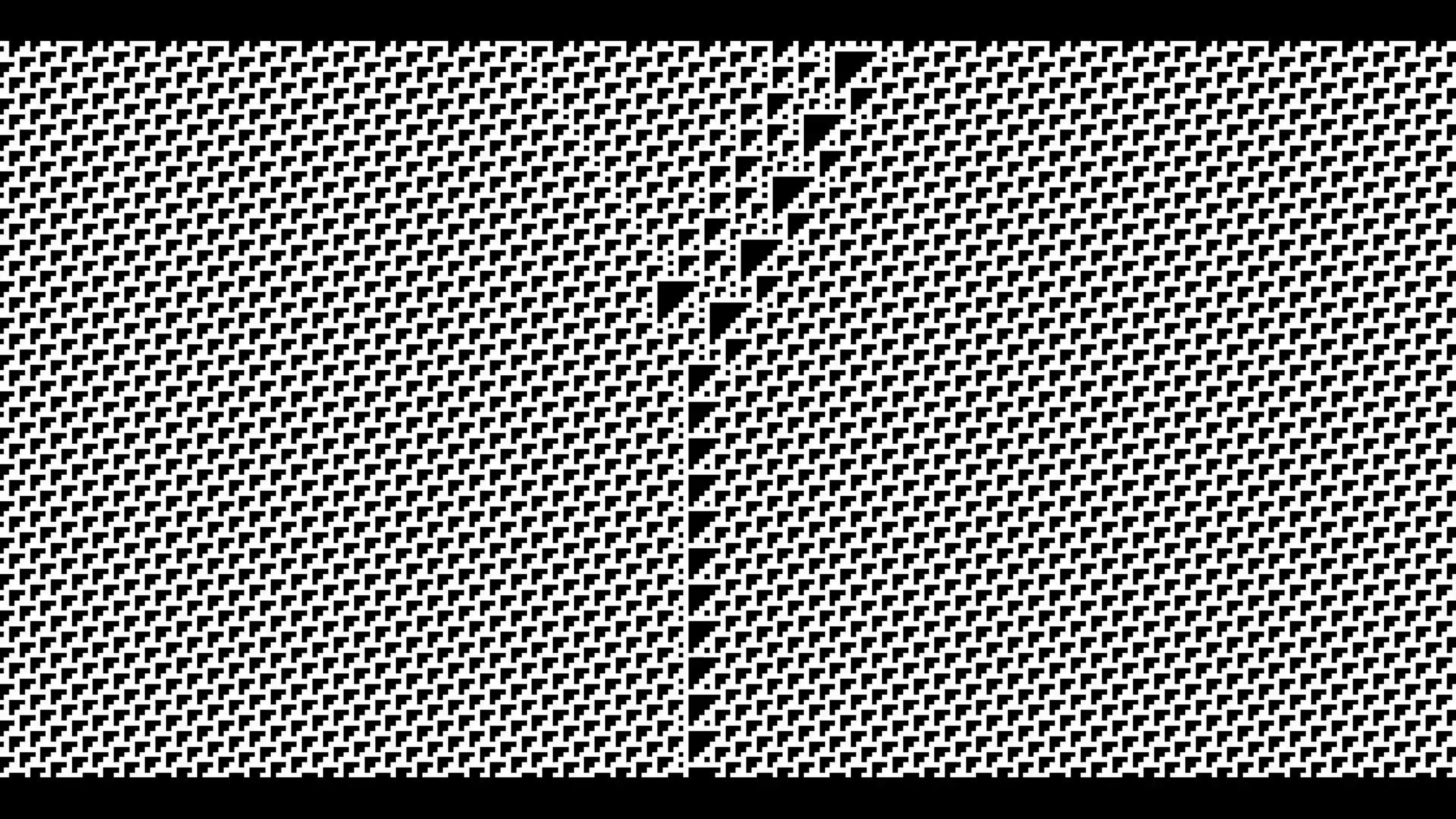


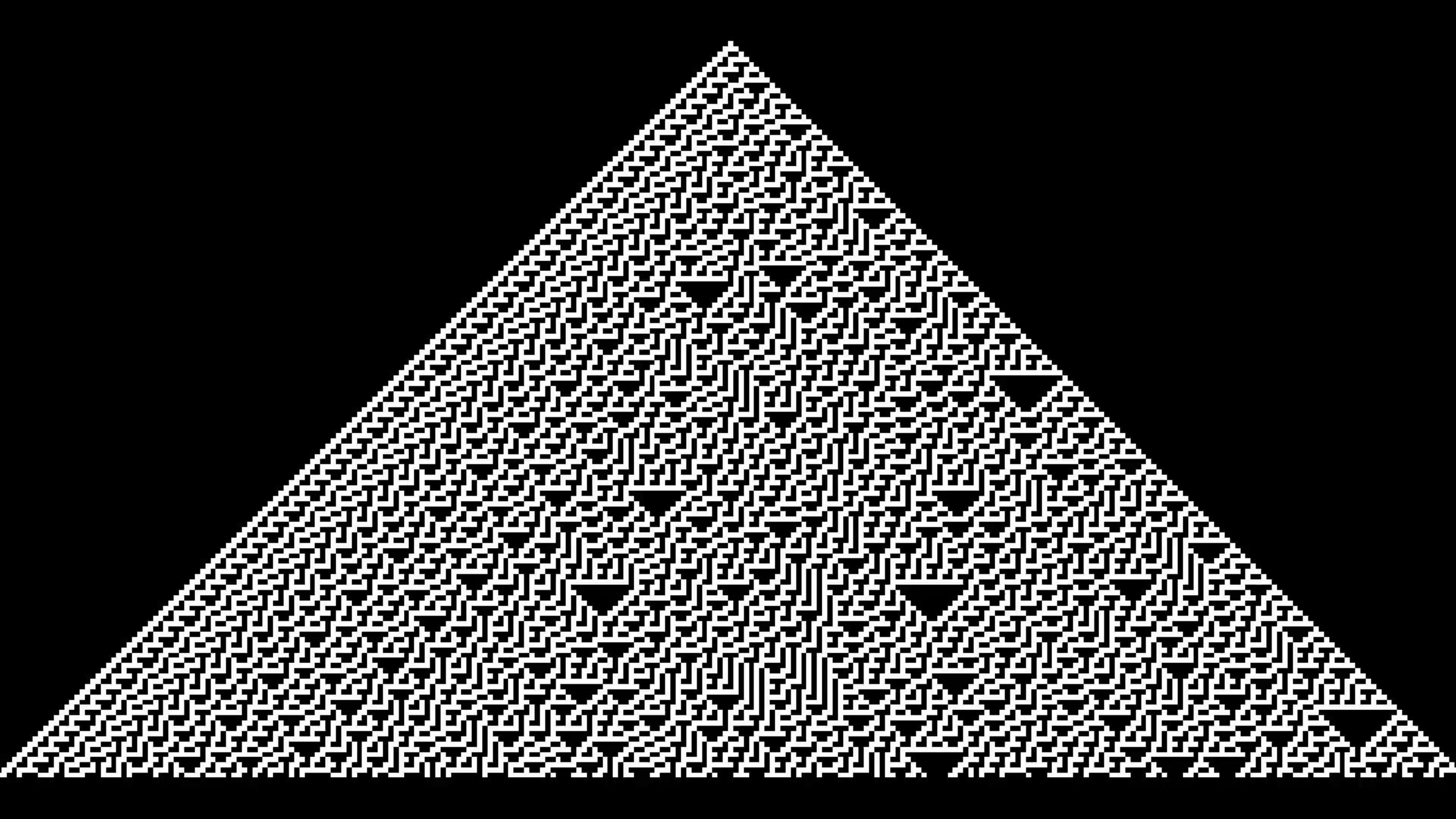


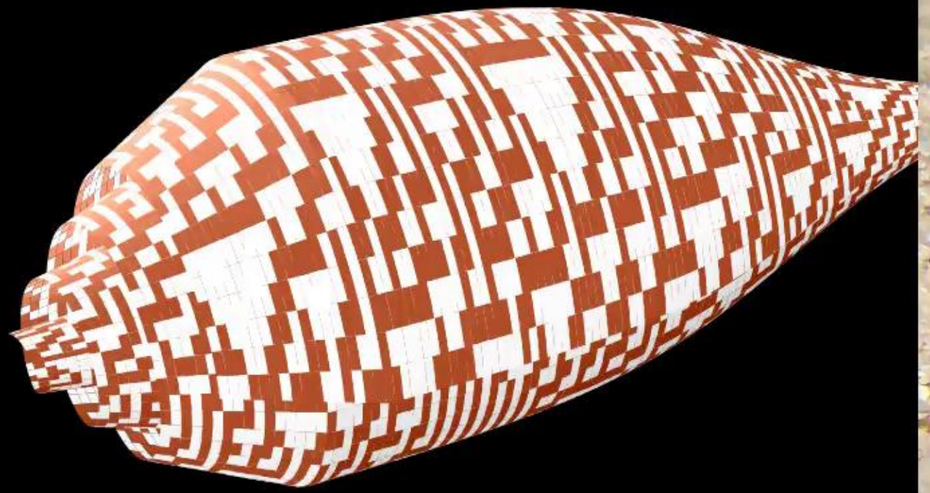




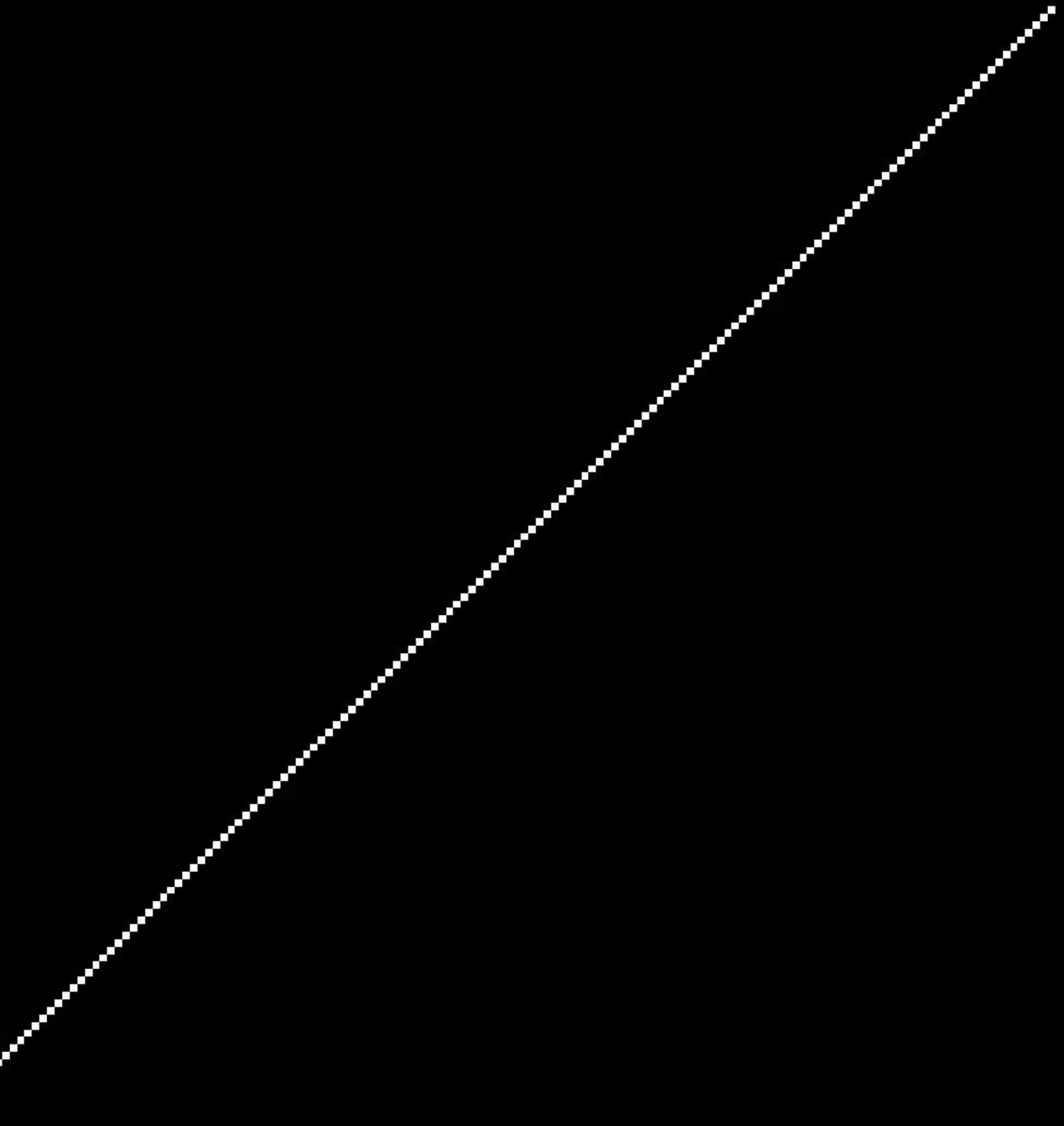














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References, media credits, and source code on Moodle